

# Pranav Minasandra, Ph.D.

Max Planck Institute of Animal Behavior,  
Bücklestraße 5a,  
78467 Konstanz, Germany

 [pminasandra \[at\] ab.mpg.de](mailto:pminasandra[at]ab.mpg.de)  
 [pminasandra.github.io](https://pminasandra.github.io)  
 [github.com/pminasandra](https://github.com/pminasandra)

**Interests** Novel and generally applicable theories of behaviour in animals; methods to explore behavioural dynamics. I am particularly interested in the dynamics of behaviour at the intersection of multiple timescales, and the *behavioural algorithms* used by animals to express individual, collective, and social behaviour.

**Experience**

|                                                                                                                                                                                                                                                                                                                                           |             |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| <b>Postdoc</b>   Max Planck Institute of Animal Behaviour, Konstanz, Germany<br>IMPRS-QBEE Postdoc Awardee.                                                                                                                                                                                                                               | 2024 - now  |
| <b>Ph.D.</b>   Max Planck Institute of Animal Behaviour, Konstanz, Germany<br><i>Magna cum laude</i><br>Thesis: “ <i>Exploring sequences of behaviour across scales and species.</i> ”<br>Also affiliated to: International Max Planck Research School of Quantitative Behaviour, Ecology, and Evolution; and the University of Konstanz. | 2020 - 2024 |
| <b>M.Sc.</b>   Indian Institute of Science (IISc), Bengaluru, India<br>Master of Science in Biology.<br>Research at the Centre for Ecological Sciences, IISc.                                                                                                                                                                             | 2019 - 2020 |
| <b>B.Sc. (Research)</b>   Indian Institute of Science (IISc), Bengaluru, India<br>Major in Biology.<br>Research at the Centre for Ecological Sciences, IISc.                                                                                                                                                                              | 2015 - 2019 |

## Publications **Peer-reviewed publications**

- Gall, G. E., Demartsev, V., **Minasandra, P.**, Baldoni, C., Cain, K. E. & Quinn, J. S. Examining Combinatoricity within the Pūkeko Vocal Repertoire. *Animal Behaviour*. 123431. doi:[10.1016/j.anbehav.2025.123431](https://doi.org/10.1016/j.anbehav.2025.123431) (2026).
- Minasandra, P.**, Grout, E. M., Brock, K., Crofoot, M. C., Demartsev, V., Gersick, A. S., Hirsch, B. T., Holekamp, K. E., Johnson-Ulrich, L., Nayak, A., *et al.* Behavioral sequences across multiple animal species in the wild share common structural features. *PNAS* **122**. e2503962122. doi:[10.1073/pnas.2503962122](https://doi.org/10.1073/pnas.2503962122) (2025). [[Press coverage](#)].
- DeNicola, V., Mezzini, S., Bursac, P., **Minasandra, P.** & Cagnacci, F. Effects of vasectomy on breeding-related movement and activity in free-ranging white-tailed deer. *Movement Ecology*. doi:[10.1186/s40462-025-00554-5](https://doi.org/10.1186/s40462-025-00554-5) (2025).
- Grout, E. M., Ortega, J., **Minasandra, P.**, Quin, M. J., Crofoot, M. C., Strandburg-Peshkin, A. & Hirsch, B. T. Whole Group Tracking Reveals That Relatedness Drives Consistent Subgrouping Patterns in White-Nosed Coatis. *Animal Behaviour* **216**. 175–193. ISSN: 0003-3472. doi:[10.1016/j.anbehav.2024.08.010](https://doi.org/10.1016/j.anbehav.2024.08.010) (2024).
- Minasandra, P.**, Jensen, F. H., Gersick, A. S., Holekamp, K. E., Strauss, E. D. & Strandburg-Peshkin, A. Accelerometer-based predictions of behaviour elucidate factors affecting the daily activity patterns of spotted hyenas. *Royal Society Open Science* **10**. 230750. doi:[10.1098/rsos.230750](https://doi.org/10.1098/rsos.230750) (2023).
- Kunjar, S., Strandburg-Peshkin, A., Giese, H., **Minasandra, P.**, Sarkar, S., Jolly, M. K. & Gradwohl, N. Link updating strategies influence consensus decisions as a function of the direction of communication. *Royal Society Open Science* **10**. 230215. doi:[10.1098/rsos.230215](https://doi.org/10.1098/rsos.230215).
- Minasandra, P.** & Isvaran, K. Truncated power-law distribution of group sizes in antelope. *Behaviour* **157**. 541–558. doi:[10.1163/1568539X-bja10012](https://doi.org/10.1163/1568539X-bja10012) (2020).

## Pre-prints and work in-prep

8. **Minasandra, P.** Anticipating others' future behaviours alters collective movement and enhances survival. *bioRxiv* **122**. 2025–08. doi:[10.1101/2025.08.14.670290](https://doi.org/10.1101/2025.08.14.670290) (2025). [In review].
9. **Minasandra\*, P.**, Sridhar\*, V. H., Roche, D. G. & Planas-Sitjà, I. TracktorLive: An Integrated Real-Time Object Tracking and Response System. *bioRxiv*. 2026.03.12.711471. doi:[10.64898/2026.03.12.711471](https://doi.org/10.64898/2026.03.12.711471) (2026). [pre-print, [GitHub](#)].
10. **Minasandra, P.**, Demartsev, V., Johnson-Ulrich, L., Manser, M. & Strandburg-Peshkin, A. Spatiotemporal interplay of vigilance, movement, and collective behaviour in groups of meerkats. [Ch. 3 of Dissertation, available [online](#)].
11. Grout, E. M., Hass, C. C., **Minasandra, P.**, Thomas, M., Ashbury, A. M., Ortega, J., Crofoot, M. C., Hirsch, B. T. & Strandburg-Peshkin, A. The vocal repertoire of white-nosed coatis: structural features, temporal dynamics, and social variations. 2026.06.17.732959. doi:[10.64898/2026.06.17.732959](https://doi.org/10.64898/2026.06.17.732959) (2026). [In review].
12. Goldsborough, Z., Crofoot, M., **Minasandra, P.**, León, J., Castillo-Caballero, P. L., del Rosario-Vargas, E. & Barrett, B. J. Habitual tool use on monopolizable resources affects group cohesion. doi:[10.32942/X25H1Q](https://doi.org/10.32942/X25H1Q) (2025). [In review].

(\* → co-first author)

## Awards and fellowships

|                                                                                                                                                                                    |             |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| <b>ISBE Travel Award</b><br>Registration fees and travel costs to ISBE conference 2026.                                                                                            | 2026        |
| <b>Best Paper Award—University of Konstanz</b><br>Chosen from among several nominations. [200 €]                                                                                   | 2025        |
| <b>Nominated—Schmidt Science Fellows programme</b><br>One of ten doctoral graduates nominated by the Germany-wide Max Planck Society.                                              | 2024        |
| <b>IMPRS Postdoctoral Award</b><br>6 months of pay at the scale of a postdoctoral researcher. [32,350 €]                                                                           | 2024 - 2025 |
| <b>ISBE Travel Award</b><br>2500 AUD [≈1,500 €].                                                                                                                                   | 2024        |
| <b>DAAD—Graduate Student Scholarship Programme</b><br>Nominated after evaluation of my proposal. [≈60,000€]                                                                        | 2020 - 2024 |
| <b>Kishore Vaigyanik Protsahan Yojana</b><br>Govt. of India scholarship for exceptional undergraduates in science. <i>All India Rank : 135</i><br>Included stipend and contingency | 2015 - 2020 |

## Mentorship

|                                                                                                                                         |  |
|-----------------------------------------------------------------------------------------------------------------------------------------|--|
| <b>Juee Dhar</b><br>Role: Mentor and Thesis Advisory Committee member. <i>Doctoral thesis</i> : Sleep social dynamics in baboon groups. |  |
| <b>Amlan Nayak</b><br>Role: Mentor and primary co-advisor. <i>Master's thesis</i> : Spatial variation of vigilance in meerkats.         |  |
| <b>Ananya Passi</b><br>Role: Co-advisor <i>Summer research</i> : Mathematical modelling of habitat use and population dynamics.         |  |

## Talks and posters

|                                                                                                                                                                                            |      |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|
| <b>[Invited] Talk at Laboratory for the Conservation of Endangered Species</b><br><i>Centre for Cellular and Molecular Biology, Hyderabad</i><br>“Three short stories in animal behaviour” | 2025 |
|                                                                                                                                                                                            | 2024 |

**[Invited] CES Seminar Series***Indian Institute of Science, Bangalore*

“Exploring sequences of behaviour in the wild across scales and species”

**International Society of Behavioural Ecology Meeting***Melbourne, Australia*

2024

“Sequences of behaviour show a common statistical structure across species”

**Gesellschaft für Primatologie Annual Conference***Konstanz, Germany*

2024

“How animals behave: Sequences of behavior show a common statistical structure across species”

**Descriptive & Normative Models of Collective Behaviour***School of Mathematics, University of Leeds*

2022

“Looking into hyena daily activity patterns using accelerometers”

**Animal Behaviour Society - Virtual Seminar 2021**

“Behavioural classifier provides insights into spotted hyena behaviour”

2021

**Teaching****Introduction to Git***Instructor and organiser:* Biannual workshop offered at the MPI of Animal Behavior.**Introduction to remote code servers like GitHub***Instructor and organiser:* Biannual workshop offered at the MPI of Animal Behavior.**L<sup>A</sup>T<sub>E</sub>X for academic documents***Instructor and organiser:* Annual workshop offered at the MPI of Animal Behavior.**Quantitative Ecology: Research Design and Inference***Teaching assistant:* Taught coding in R, conducted several classes, graded assignments, managed course website.

2019

**Technical  
skills***Programming languages and related*Python, Julia, R, Bash, L<sup>A</sup>T<sub>E</sub>X, and C. HTML, CSS, and git. Familiarity with agentic code generation.*Specialised computational skills*

Machine learning; Parallel processing and Inter-Process Communication; Agent-based models; Analysis of complex multivariate time-series data.

*Mathematical modelling*

Dynamical (PDE) models incorporating spatial and stochastic variables; Probability models; Numerical simulations of all the above.

**Other  
academic  
services****Reviewing**

Methods in Ecology and Evolution; PLOS One; Integrative and Comparative Biology; Animal Behaviour.

**Representation**

Postdoc representative at the Max Planck Institute of Animal Behaviour.

2024 - 2025

**References****Dr Ariana Strandburg-Peshkin,**Max Planck Institute of Animal Behavior  
astrandburg@ab.mpg.de**Dr Kavita Isvaran,**Indian Institute of Science  
kavita@iisc.ac.in**Dr Andrew Berdahl,**University of Washington  
berdahl@uw.edu**Prof Meg Crofoot,**Max Planck Institute of Animal Behavior  
mcrofoot@ab.mpg.de